PROJECT EVALUATION REPORT

MoE based customer segmentation model

Hierarchical mixture-of-experts for behavioral, text and temporal customer views

0.590873 Hierarchical MoE silhouette	**+0.032945** Gain vs best static fusion	**0.999912** Stability ARI

CENTRAL FINDING

Customer-specific adaptive routing can outperform strong static fusion, but only after the routing design is strengthened. The hierarchical MoE is the strongest multimodal routing result; its downstream retention value remains to be validated.

Brazilian E-Commerce Public Dataset (Olist)
Research cohort: 57,510 customers
Analysis date: 25 October 2026 • Status: Phase 13 complete

Executive summary

Decision takeaway
Advance the hierarchical MoE to downstream and robustness validation as a **champion multimodal segmentation candidate**. Do not yet treat it as a production replacement or as proven to improve retention: Phase 13 retention results are not yet reported.

This report evaluates whether customer-specific adaptive routing across behavioral, text and temporal views improves customer segmentation over strong unimodal and static-fusion baselines. The analysis uses a 57,510-customer Olist cohort and compares internal cluster geometry, assignment stability and downstream retention evidence.

Headline measure	Result	Business interpretation
Hierarchical MoE geometry	**Silhouette 0.590873**	Best Phase 13 multimodal routing result
Gain over best static fusion	**+0.032945** (+5.9% relative)	Exceeds text + temporal fixed-weight fusion at 0.557928
Other geometry metrics	**DBI 0.638034; CH 19,223.176**	Lower DBI and higher CH than the best static baseline
Assignment stability	**ARI 0.999912**	Extremely high repeated-fit stability in the reported evaluation
Downstream retention	**Not yet evaluated for Phase 13**	Business utility remains an open validation requirement

What the evidence supports

- **Adaptive routing is conditionally successful.** Simple learned gates failed in Phase 10; the hierarchical design succeeded against the strongest static-fusion reference in Phase 13.
- **The result is not a universal geometry victory.** Temporal-only silhouette is 0.596058, just 0.005185 above hierarchical MoE. Hierarchical MoE nevertheless reports slightly better DBI and higher CH than temporal-only.
- **More modalities are not automatically better.** Several static concatenation and full three-view variants underperform temporal-only and text + temporal fusion.
- **Availability information matters.** Removing the text-availability indicator reduces text + temporal silhouette from 0.557928 to 0.319270 and stability ARI from 1.000000 to 0.862250.
- **RFM remains the retention leader.** KMeans RFM delivers 3.756427× corrected retention lift, with 4.705882% maximum cluster retention against a 1.252755% test baseline.

Recommended decision

Use a **champion–challenger validation strategy**: hierarchical MoE as the multimodal champion, temporal-only and best static fusion as geometry challengers, and KMeans RFM as the interpretable retention benchmark. Production selection should depend on downstream retention, segment actionability and fair same-cohort comparisons—not silhouette alone.

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How to read this report

- **Geometry** describes how compact and separated the clusters are; it does not prove that segments are commercially useful.
- **Stability** describes reproducibility of assignments under the reported procedure; it is not out-of-sample accuracy.
- **Retention lift** is a downstream signal based on rare future events and should be read with confidence intervals and statistical-power limitations.
- **Cross-phase comparisons are directional unless the cohort, features, dimensionality, K and evaluation protocol are aligned.** This is especially important for Phase 6 and text-only results.

Scope and provenance
All numerical results come from the supplied project analysis. Derived differences and percentages are calculated from those values. Repository structure, artifact inventories, file paths and reproduction commands are intentionally excluded.

1 Research context and data foundation

1.1 Research question

The central hypothesis is that **customer-specific adaptive routing** can combine behavioral, text and temporal evidence more effectively than fixed-weight fusion. Instead of assigning the same modality weights to every customer, a mixture-of-experts model learns which views should influence each customer representation.

> **Hypothesis verdict: conditionally supported**
> Phase 10 rejects the hypothesis for simple learned gates. Phase 13 supports it for the improved hierarchical architecture when compared with the best static-fusion baseline. The evidence does not yet establish superior retention or superiority over every unimodal configuration.

1.2 Cohort and observation design

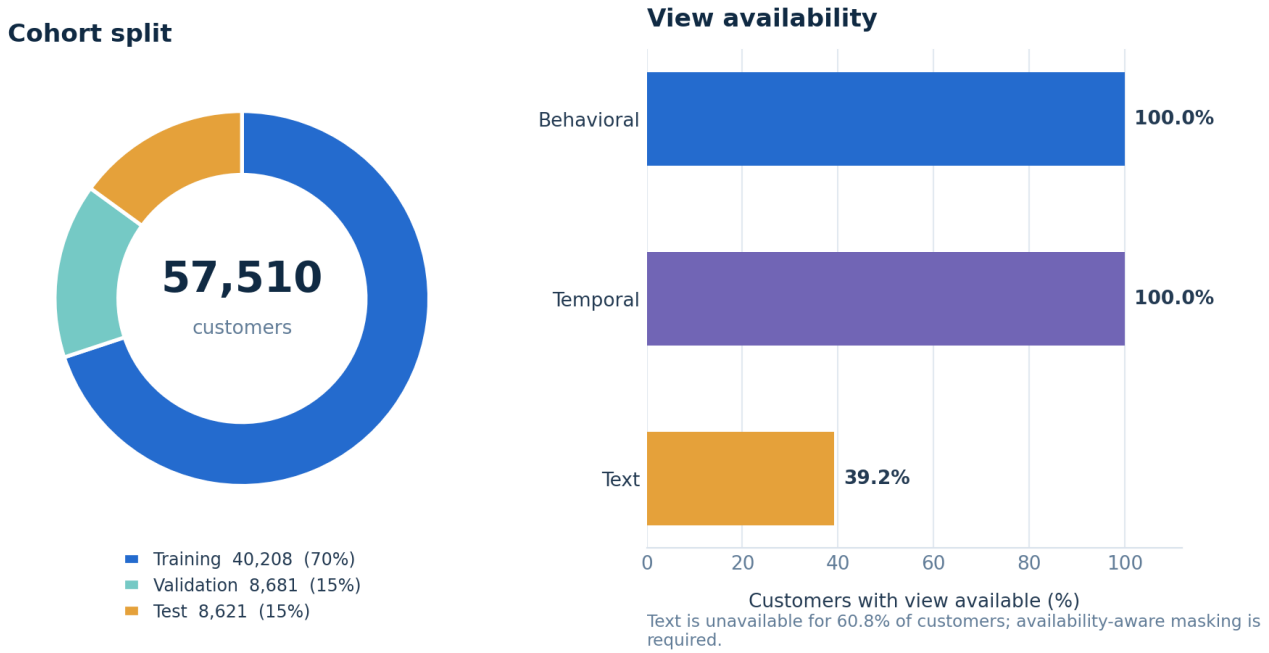


Figure 1. Research cohort split and availability of the three customer views.

Design element	Reported value	Purpose
Research cohort	**57,510 customers**	Common customer population for Phases 5–13
Training / validation / test	40,208 / 8,681 / 8,621	Approximately 70% / 15% / 15%
Observation cutoff	**1 March 2018**	Only pre-cutoff information may inform representations
Future evaluation window	1 March–1 September 2018	Six-month downstream retention evaluation
Test retention baseline	108 / 8,621 = **1.252755%**	Reference for corrected retention lift
Cohort retention baseline	684 / 57,510 = 1.189358%	Overall future-event rate

1.3 Customer views and representation coverage

View	Availability	Source information	Representation used for fusion
Behavioral	**57,510 (100%)**	RFM, logistics, payment and satisfaction features	8 selected non-redundant features → 32 dimensions
Text	**22,535 (39.2%)**	Eligible non-empty customer reviews before cutoff	384-dimensional MiniLM embedding → 32 dimensions
Temporal	**57,510 (100%)**	Pre-cutoff customer order-event sequences	GRU autoencoder embedding → 32 dimensions

Behavioral features cover purchase frequency and value, product and seller diversity, recency and lifetime, delivery performance, payment behavior and review-based satisfaction. The text view is sparse: **60.8% of customers have no usable text representation**. Temporal sequences are available for every customer, but 96.88% have only one order, limiting sequence depth for most of the cohort.

2 Modeling approach and evaluation framework

2.1 From source data to customer representations

The modeling pipeline converts raw e-commerce tables into validated customer-level analytical views, then applies train-fitted preprocessing and expert-specific representations. The design separates representation learning from downstream retention labels so that future behavior is used only for evaluation.

Stage	Transformation	Modeling safeguard
Source ingestion	Schema typing, datetime parsing and lossless storage	Raw source data remains unchanged
Entity normalization	Customer, order, product and review normalization with quality flags	Consistent customer grain and deduplication
Analytical views	Customer profiles, order events, reviews and category preferences	One aligned customer identifier across views
Expert processing	Imputation, scaling, PCA projection and view-specific embeddings	Scaler and PCA fitted on training customers only
Routing and clustering	Availability-masked expert fusion followed by segmentation	Geometry, stability and retention evaluated separately

2.2 Expert representations

Expert	Input	Processing	Output
Behavioral	8 selected features	Train-fitted StandardScaler + PCA	32 dimensions
Text	384-dimensional review embedding	all-MiniLM-L6-v2; mean pooling; train-fitted scaling + PCA	32 dimensions
Temporal	Pre-cutoff event sequences; maximum length 20	GRU autoencoder; later-known fields excluded; train-fitted scaling + PCA	32 dimensions

Text preprocessing used 54,778 eligible reviews, of which 22,973 were non-empty, producing representations for 22,535 customers. Temporal processing used 59,427 pre-cutoff events and produced finite embeddings for all 57,510 customers. No cutoff violations or non-finite customer embeddings were reported.

2.3 Hierarchical routing design

The improved model replaces a single flat gate with a **two-level routing hierarchy**. The coarse gate weighs behavioral evidence against a combined text + temporal path; the fine gate allocates weight between text and temporal evidence. Unavailable views are masked and the remaining weights are renormalized.

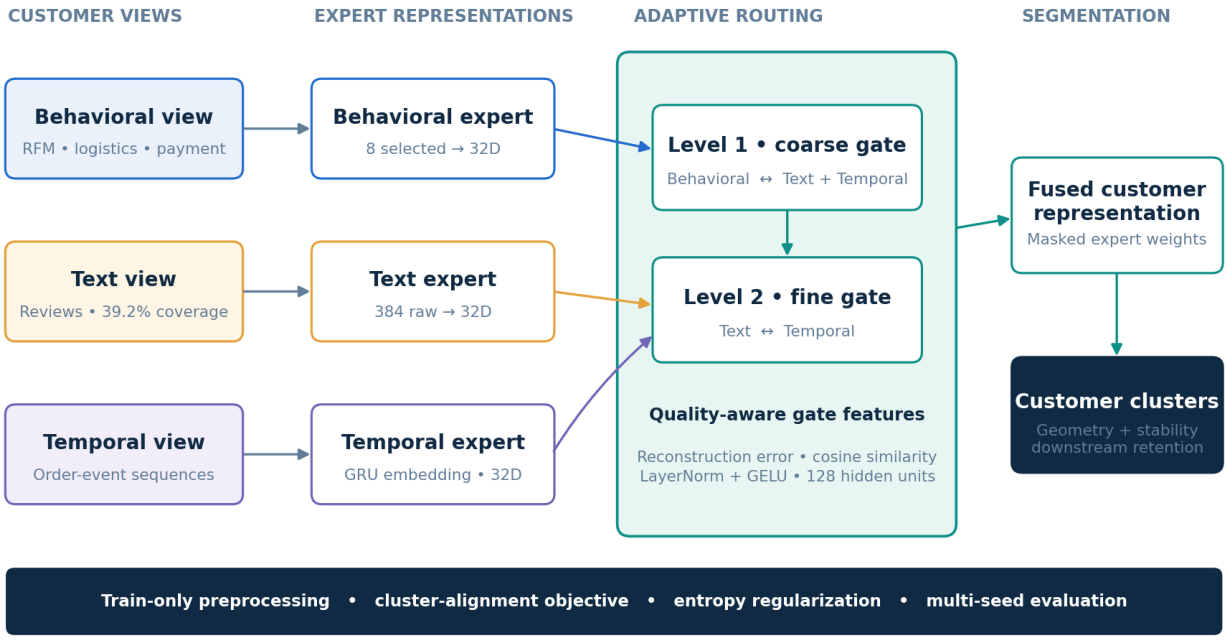


Figure 2. Conceptual architecture of the hierarchical mixture-of-experts segmentation model.

2.4 Phase 13 routing enhancements

Enhancement	Purpose	Evidence role
Hierarchical gating	Separate coarse and fine routing decisions	Defining feature of the best Phase 13 variant
DEC cluster-alignment objective	Optimize routing toward separated cluster structure	Addresses the weak reconstruction-only objective used earlier
Residual gate + LayerNorm + GELU	Increase gate capacity and improve optimization	128 hidden units versus 24 in Phase 10
Quality-aware gate features	Expose reconstruction error and pairwise cosine similarity	Provides information about view reliability and agreement
Longer training + cosine learning rate	Improve convergence	100 epochs versus 10 in Phase 10
Entropy and sparse-routing mechanisms	Discourage degenerate routing and test explicit modality selection	Sparse top-1 evaluated as a distinct Phase 13 variant
Five-seed ensembling	Reduce sensitivity to initialization	Supports robustness of the final routing result

Attribution note

Phase 13 demonstrates the success of the improved design package. It does not provide a complete component-by-component ablation of the final hierarchical model, so the marginal contribution of each enhancement should not be assumed.

2.5 Evaluation framework

Measure	Direction	What it reveals	Important caveat
Silhouette	Higher is better	Cluster cohesion relative to separation	Sensitive to representation, distance and cohort
Davies–Bouldin index (DBI)	Lower is better	Average within-cluster to between-cluster similarity	Should be interpreted with other geometry metrics
Calinski–Harabasz (CH)	Higher is better	Between-cluster dispersion relative to within-cluster dispersion	Scale depends on sample size and feature geometry
Stability ARI	Higher is better	Agreement of assignments across repeated fits	Not a measure of external validity or prediction accuracy
Corrected retention lift	Higher is better	Maximum cluster retention relative to the test baseline	Rare events create uncertainty and wide intervals
Omnibus p-value	Smaller indicates stronger evidence	Overall difference in retention across clusters	Exploratory test-set use and multiplicity limit confirmatory interpretation

2.6 Leakage prevention and data quality

- Review text and temporal events are restricted strictly to dates before the observation cutoff; **zero violations** were reported.
- Delivery duration, review scores and final statuses are excluded from temporal inputs when they would only become known later.
- Scaling and PCA are fitted on training customers only; future retention labels are not used for model fitting.
- Customer rows are unique, embeddings are finite and assignment alignment is maintained across the relevant phases.
- Earlier exploration of the blind test set means the test partition should not be described as a pristine, untouched holdout.

3 Legacy baselines and retention evidence

Legacy baselines establish two separate reference points: **cluster geometry** and **downstream retention separation**. KMeans RFM is the strongest overall legacy model because it combines good geometry with the highest reported retention lift.

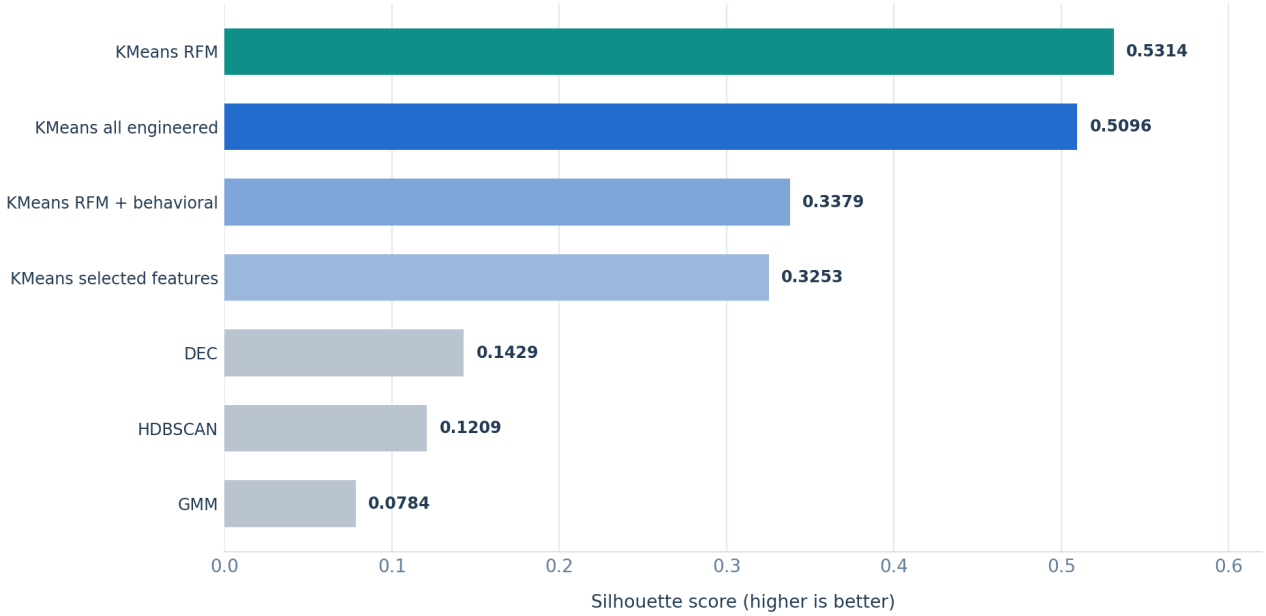


Figure 3. Silhouette scores for the legacy baseline models. Phase 6 uses a different protocol from the later fusion experiments.

Table 1. Selected Phase 6 geometry and corrected retention metrics.

Model	K	Silhouette	DBI	Corrected retention lift	Omnibus p
KMeans RFM	5	**0.5314**	**0.6182**	**3.756427×**	**<0.00001**
KMeans all engineered	4	0.5096	1.1484	3.583938×	<0.0001
KMeans RFM + behavioral	4	0.3379	1.0759	3.456948×	<0.00001
KMeans selected features	5	0.3253	1.2518	3.712748×	<0.0001
DEC	5	0.1429	1.6039	3.712748×	<0.0001
HDBSCAN	6	0.1209	1.9047	2.073353×	0.45444
GMM	7	0.0784	1.9130	2.673534×	0.00833

3.1 Retention separation

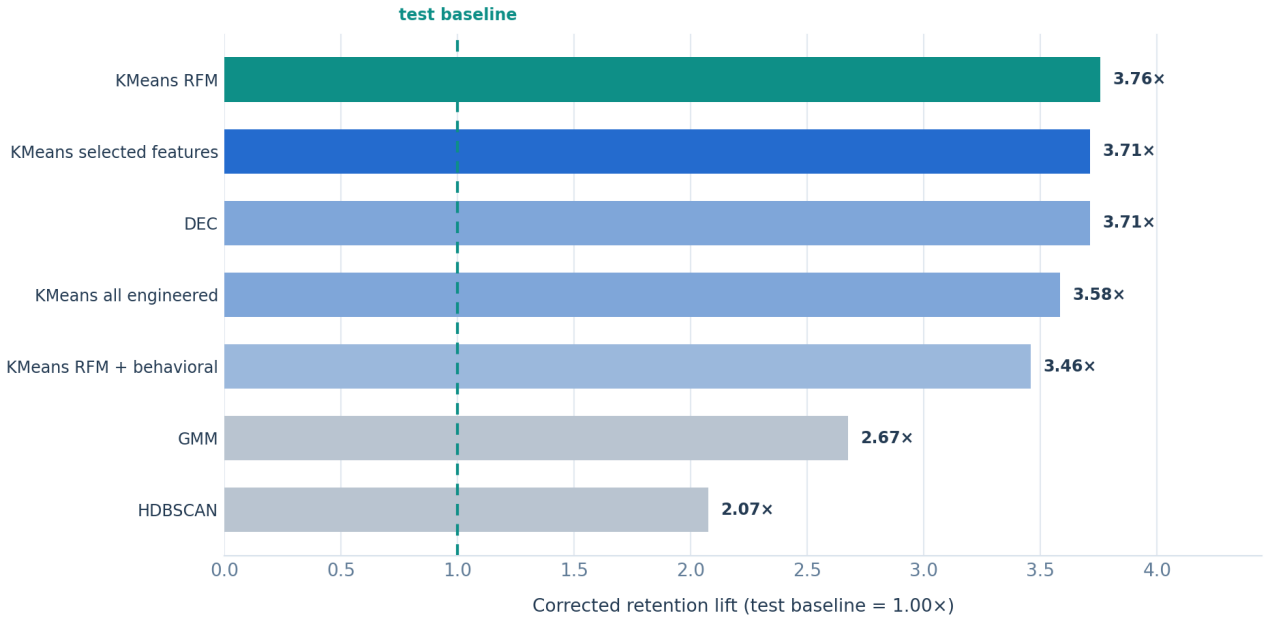


Figure 4. Corrected retention lift for legacy models, using the test retention rate as the 1.00× reference.

Strongest downstream result: KMeans RFM
The highest-retention RFM cluster reaches **4.705882%** retention versus the **1.252755%** test baseline, producing **3.756427× corrected lift**. The Wilson 95% interval is 2.712159%–8.044023%, Cohen's h is 0.213015, and the omnibus test is significant at $p < 0.00001$.

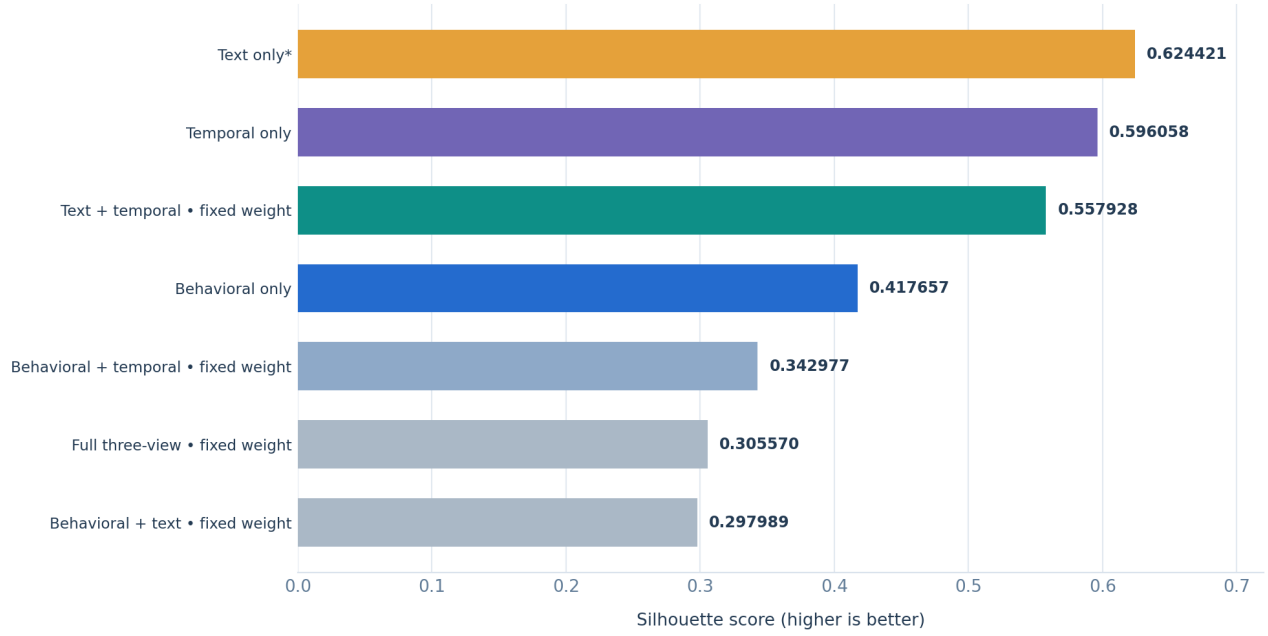
The retention results also show why geometry alone is insufficient for model selection. HDBSCAN has perfect repeated-fit stability in the saved evaluation but no significant retention separation ($p = 0.45444$). Conversely, RFM's silhouette is not the highest reported across all experiments, yet it provides the clearest downstream retention signal among models with reported retention results.

Comparability note: Phase 6 uses different feature sets, dimensionality, K values and evaluation procedures from Phases 9–13. These results should be used as reference points, not as a single perfectly controlled leaderboard.

4 Static fusion and initial MoE routing

4.1 Static multimodal fusion

Static fusion applies the same combination rule to every customer. It is an essential benchmark because an adaptive model only adds value if it can outperform a strong non-adaptive alternative. The strongest static-fusion result is **text + temporal fixed-weight fusion**, with silhouette 0.557928 and stability ARI 1.000000.



* Text-only uses the text-available cohort; do not treat its score as a like-for-like full-cohort ranking.

Figure 5. Selected unimodal and projected fixed-weight fusion results. Text-only is shown with a cohort-comparability caveat.

Variant	Dimensions	Silhouette	DBI	CH	Stability ARI
Text only*	385	0.624421	3.474894	1,704.446	0.990473
Temporal only	32	0.596058	0.663542	18,176.656	1.000000
Text + temporal fixed weight	34	**0.557928**	**0.779689**	**9,113.616**	**1.000000**
Behavioral only	8	0.417657	1.157549	3,857.446	0.999362
Behavioral + temporal fixed weight	34	0.342977	1.128105	4,970.821	0.994900
Full three-view fixed weight	35	0.305570	1.245361	6,564.488	0.998652
Behavioral + text fixed weight	34	0.297989	1.451636	3,872.340	0.998484

* The text-only configuration is associated with the text-available cohort and has a high DBI; its silhouette should not be interpreted as conclusive full-cohort superiority. The full static results also show that direct concatenation can degrade geometry: behavioral + text + temporal concatenation reports silhouette 0.206452.

4.2 Availability-aware masking is essential

Phase 12 tests whether text adds reproducible value beyond temporal geometry when the availability indicator is removed. The decision rule requires both silhouette and stability to improve. Instead, **both deteriorate sharply**.

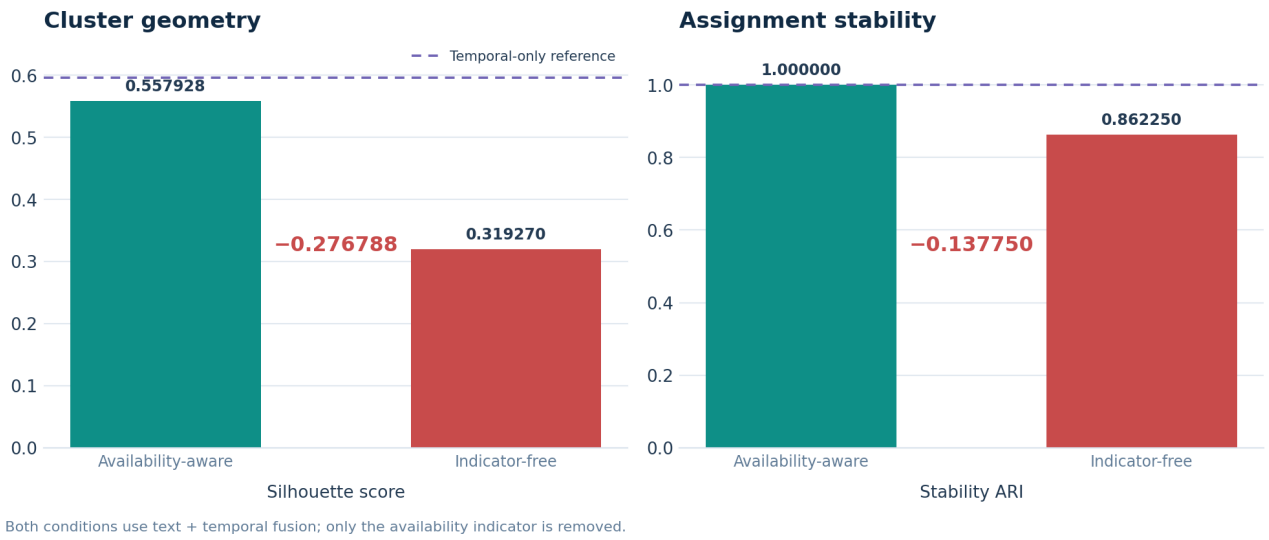


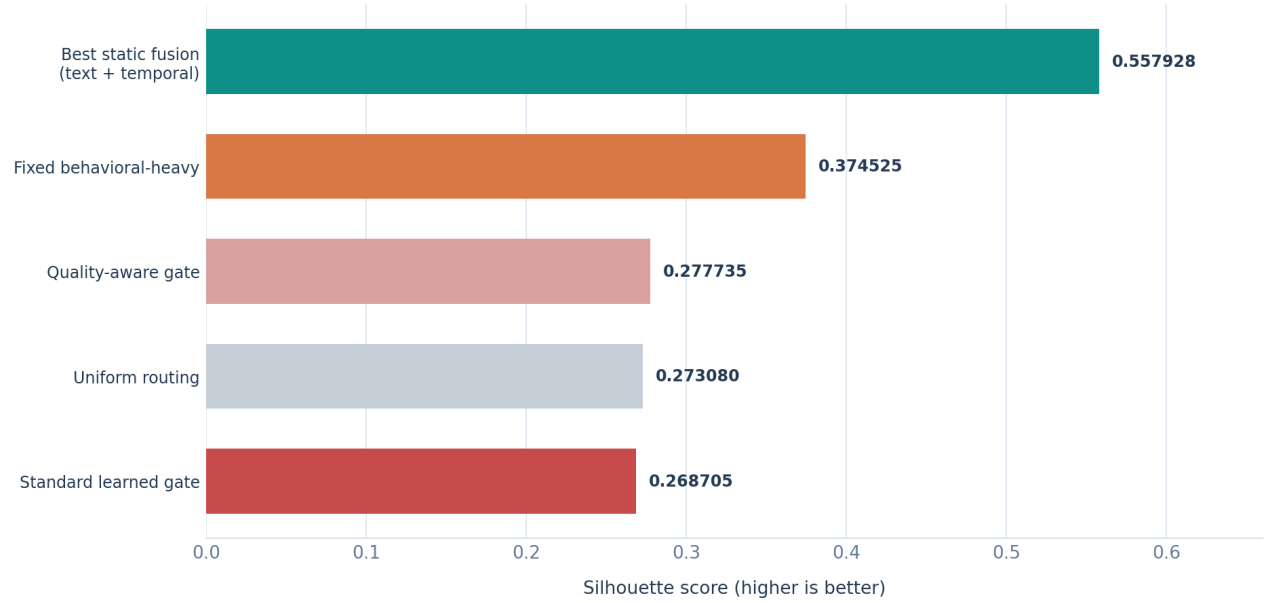
Figure 6. Effect of removing the availability indicator from text + temporal fusion.

Variant	Silhouette	Δ vs temporal only	Stability ARI	Decision
Temporal only	0.596058	0.000000	1.000000	Reference
Text + temporal • availability-aware	0.557928	−0.038130	1.000000	Strong static benchmark
Text + temporal • indicator-free	0.319270	−0.276788	0.862250	**Fails decision rule**

Design implication
Missing text must be represented explicitly. Mask unavailable experts, renormalize weights over available views and preserve availability information. The report does not support the claim that text independently improves temporal-only geometry under the indicator-free test.

4.3 Phase 10: simple adaptive routing underperforms

The initial MoE compared uniform routing, fixed behavioral-heavy weights, a standard learned gate and a quality-aware gate. The best MoE result—fixed behavioral-heavy routing—achieved silhouette 0.374525, **0.183403 below** the best static-fusion baseline.



Best Phase 10 MoE result trails the best static fusion baseline by 0.183403 silhouette points.

Figure 7. Phase 10 routing variants compared with the strongest static-fusion baseline.

Routing variant	Silhouette	DBI	Stability ARI	Single-view rate	Mean entropy
Uniform	0.273080	1.290308	0.995613	0.000000	0.852027
Fixed behavioral-heavy	**0.374525**	**1.214152**	**0.999690**	0.000000	0.714346
Standard learned gate	0.268705	1.345548	0.978325	0.000000	0.850875
Quality-aware gate	0.277735	1.336678	0.997184	0.000000	0.850666

No Phase 10 routing variant selected a single view, and the learned gates did not materially improve on uniform routing. This indicates that merely learning customer-specific weights is insufficient: the training objective, gate capacity and routing structure must encourage useful cluster specialization.

4.4 Ablation evidence: geometry and retention diverge

Ablation	Silhouette	Retention omnibus p	Interpretation
Behavioral only	0.417658	0.099657	No significant overall retention separation at 5%
Temporal only	0.596058	0.664322	Strong geometry; weak retention evidence
Text + temporal fixed weight	0.557928	0.507282	Strong static geometry; weak retention evidence
Fixed behavioral-heavy	0.374525	**0.037501**	Only listed ablation with significant retention separation
Standard learned gate	0.268705	0.377040	No significant retention separation
Quality-aware gate	0.274782	0.387313	No significant retention separation

Phase 10 verdict
The initial adaptive-routing hypothesis is **rejected for simple learned gates**. The ablations also warn against equating high silhouette or stability with downstream customer value.

5 Hierarchical MoE results

Phase 13 revisits adaptive routing with a deeper gate, quality-aware features, cluster-aligned training, longer optimization, entropy and sparsity mechanisms, hierarchical routing and multi-seed evaluation. The **hierarchical MoE** is the only listed Phase 13 variant to exceed the best static-fusion silhouette.

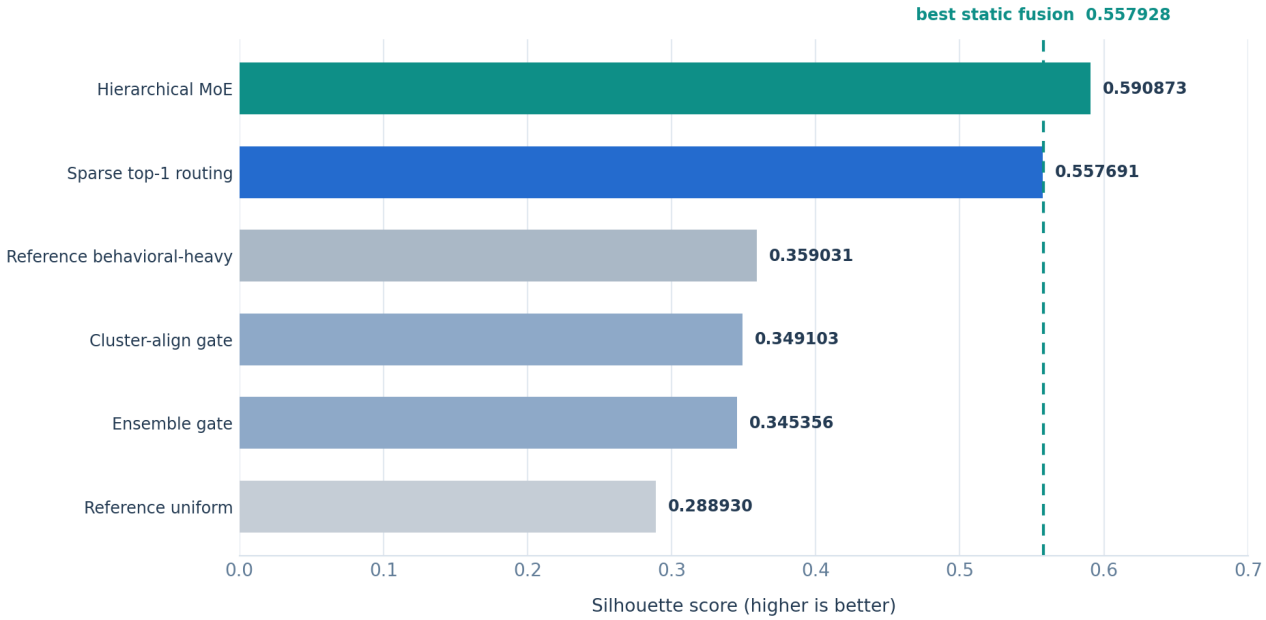


Figure 8. Phase 13 routing variants. The dashed line marks the best static-fusion result from Phase 9.

Table 2. Phase 13 routing performance with the Phase 9 static reference.

Variant	Silhouette	DBI	CH	Stability ARI
Hierarchical MoE	**0.590873**	**0.638034**	**19,223.176**	**0.999912**
Sparse top-1 routing	0.557691	0.703640	17,210.058	0.999592
Reference behavioral-heavy	0.359031	1.259050	3,470.597	0.999397
Cluster-align gate	0.349103	1.039966	10,079.353	0.999843
Ensemble gate	0.345356	1.023713	9,218.411	0.989008
Reference uniform	0.288930	1.258587	4,315.472	0.999292
Best static fusion (reference)	0.557928	0.779689	9,113.616	1.000000

Sparse top-1 routing is effectively tied with the best static baseline on silhouette (−0.000237), while improving DBI and CH. The cluster-align and ensemble variants improve some diagnostics but do not exceed static fusion on silhouette. The hierarchy is therefore central to the headline result, although a full factorial ablation is still needed for definitive attribution.

5.1 Headline comparison

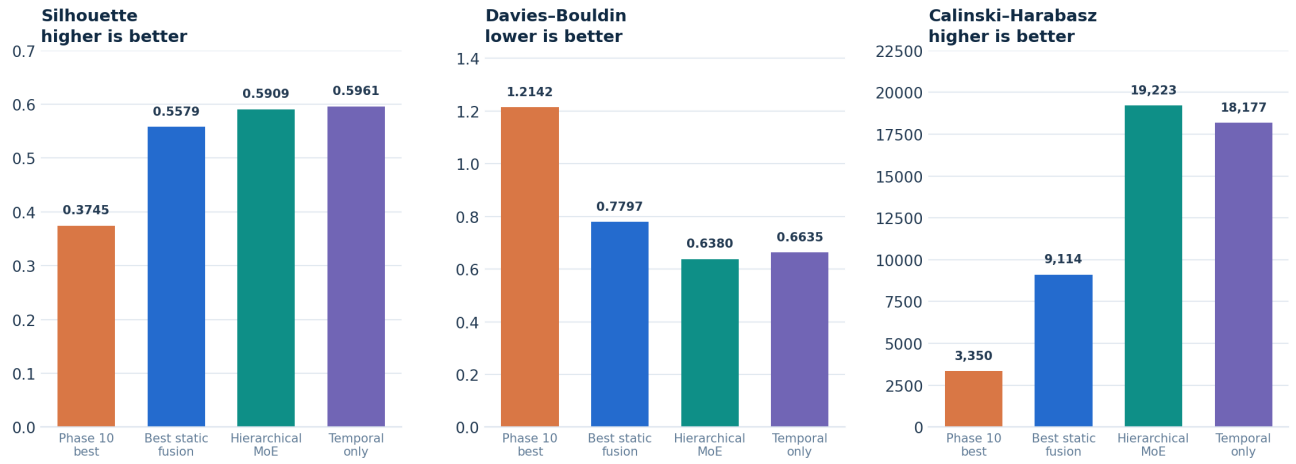


Figure 9. Geometry comparison across the initial MoE, best static fusion, hierarchical MoE and temporal-only references.

Model	Silhouette	DBI	CH	Stability ARI	Evidence-based interpretation
Phase 10 best MoE	0.374525	1.214152	3,350.098	0.999690	Simple routing is insufficient
Best static fusion	0.557928	0.779689	9,113.616	1.000000	Strong non-adaptive multimodal benchmark
Hierarchical MoE	**0.590873**	**0.638034**	**19,223.176**	**0.999912**	**Best Phase 13 multimodal routing result**
Temporal only	0.596058	0.663542	18,176.656	1.000000	Close unimodal geometry benchmark

5.2 Improvement over the initial MoE and static fusion

Metric	Phase 10 best	Best static fusion	Hierarchical MoE	Hierarchical MoE change
Silhouette	0.374525	0.557928	**0.590873**	+0.216348 vs Phase 10; **+0.032945 vs static**
DBI	1.214152	0.779689	**0.638034**	−0.576118 vs Phase 10; −0.141655 vs static
CH	3,350.098	9,113.616	**19,223.176**	+15,873.078 vs Phase 10; +10,109.560 vs static
Stability ARI	0.999690	1.000000	**0.999912**	+0.000222 vs Phase 10; −0.000088 vs static

Against best static fusion, hierarchical MoE improves silhouette by **5.9% in relative terms**, reduces DBI by **18.2%** and increases CH by **110.9%**. Its stability ARI remains effectively perfect, although it is marginally below the static and temporal references at 1.000000.

Precise Phase 13 conclusion
The hierarchical MoE **does outperform the strongest static-fusion baseline** on silhouette, DBI and CH. It remains **0.005185 below temporal-only silhouette**, while reporting better DBI and higher CH than temporal-only. The most defensible claim is therefore **superior multimodal routing with competitive overall geometry**, not unconditional dominance over every model.

5.3 What remains unproven

- **Retention utility:** no Phase 13 hierarchical MoE retention result is reported.
- **Component attribution:** the final hierarchy has not been subjected to a complete leave-one-enhancement-out analysis.
- **Robustness and actionability:** high stability is not external validity, and original-feature cluster profiles or treatment recommendations are not included in the supplied results.

6 Business interpretation and model selection

The appropriate model depends on the decision being supported. Internal geometry favors a different shortlist than retention-focused campaign targeting, and an interpretable baseline may remain valuable even when a more complex model achieves better geometric separation.

6.1 Recommended model by use case

Use case	Recommended model	Why	Condition or caveat
Multimodal segmentation champion	**Hierarchical MoE**	Best Phase 13 multimodal geometry; 0.590873 silhouette and strong DBI / CH	Validate retention, actionability and robustness before rollout
Simple geometry benchmark	**Temporal only**	Silhouette 0.596058; excellent stability and simple 32-dimensional view	Short sequences for 96.88% single-order customers limit behavioral depth
Non-adaptive fusion challenger	**Text + temporal fixed weight**	Strong static baseline with 0.557928 silhouette and ARI 1.000000	Requires availability-aware masking
Retention-focused campaigns	**KMeans RFM**	Highest corrected retention lift: 3.756427×; simple and interpretable	Different Phase 6 protocol; validate on aligned cohorts
Flat learned routing	**Not recommended**	Phase 10 gates underperform static fusion	Do not deploy without architectural and objective improvements

6.2 Geometry is not a substitute for business value

The project contains two strong but different success stories. **Hierarchical MoE provides the most compelling multimodal geometry result**, while **KMeans RFM provides the strongest reported retention separation**. Because Phase 13 retention is pending, it is not yet valid to claim that the more complex model creates greater marketing or customer-lifetime value.

> **Current evidence leaders**
> **Hierarchical MoE** leads multimodal routing geometry; **KMeans RFM** leads reported retention lift; **text + temporal fixed weight** remains the static-fusion challenger. Phase 13 retention and segment treatments are not reported, so downstream superiority and intervention design remain open.

6.3 Deployment recommendation

- **Run the Phase 13 retention evaluation.** Use the same retention definition and corrected-lift calculation as the baseline evaluation; report cluster sizes, Wilson intervals, effect sizes and omnibus results.
- **Profile every hierarchical MoE cluster.** Describe recency, frequency, monetary value, category mix, satisfaction, logistics experience, text availability and temporal behavior in the original feature space.
- **Perform a fair champion–challenger comparison.** Align cohort, preprocessing, K, distance metric and evaluation protocol across hierarchical MoE, temporal-only, static fusion and RFM.
- **Validate actionability.** Assign a specific business question or intervention to each segment and test whether the segmentation changes a decision, not merely a metric.
- **Monitor routing behavior.** Track gate weights, entropy, selected experts, missing-view rates, cluster sizes, stability and drift by customer subgroup.
- **Roll out only after downstream and robustness gates pass.** Until then, treat hierarchical MoE as a validated research candidate rather than a production default.

7 Limitations and validation roadmap

7.1 Limitations that affect interpretation

Limitation or risk	Why it matters	Required mitigation
Cross-phase comparability	Phase 6 and later phases differ in features, dimensionality, K and protocol	Run an aligned same-cohort benchmark before declaring an overall winner
Earlier test-set exploration	The test partition is not a pristine untouched holdout	Use a new temporal holdout or external validation set for confirmation
Rare retention events	Only 108 retained test customers; some cluster intervals are wide	Report uncertainty, avoid overinterpreting small differences and seek larger event counts
Phase 13 retention is pending	Geometry gains may not translate into downstream value	Complete retention and actionability evaluation before deployment
Text coverage is only 39.2%	Text-only comparisons use a different subpopulation; missingness can encode information	Use explicit masks and report performance by text-availability subgroup
96.88% single-order customers	Most temporal sequences are very short	Test simpler temporal features and assess performance by order-history depth
HDBSCAN is transductive in the saved evaluation	Perfect repeated-fit stability is not out-of-sample predictive stability	Separate transductive consistency from inductive validation
Component attribution is incomplete	Phase 13 changes several design elements together	Run leave-one-component-out and interaction ablations around the final hierarchy
Internal metrics do not measure treatment value	Compact clusters may still be commercially meaningless	Evaluate segment profiles, intervention lift and operational usability

7.2 Prioritized validation roadmap

Priority	Workstream	Analysis required	Decision gate
1	**Downstream utility**	Phase 13 retention, cluster sizes, confidence intervals and effect sizes	Evidence of meaningful, statistically credible separation
2	**Fair benchmarking**	Aligned cohort / features / K / protocol across key challengers	Hierarchical MoE retains its advantage under controlled comparison
3	**Segment actionability**	Original-feature profiles, gate-weight patterns and treatment hypotheses	Segments are interpretable and connected to decisions
4	**Robustness**	New temporal holdout, seed sensitivity, subgroup and missingness analysis	No material fragility across time, seeds or availability groups
5	**Component ablation**	Hierarchy, cluster loss, quality features, sparsity and ensembling	Identify the simplest design that preserves the gain
6	**Operational readiness**	Inference pipeline, monitoring, fallback routing and retraining policy	Reliable serving with explicit degradation behavior

7.3 Suggested next-phase analysis design

- **Aligned comparisons:** evaluate hierarchical MoE against static fusion and temporal-only on identical customers and clustering settings; reconcile the Phase 6 protocol before comparing retention with RFM.
- **Subgroup and uncertainty analysis:** compare text-available versus text-missing and single-order versus repeat-order customers, using repeated temporal validation or bootstrapping where feasible.
- **Simplification analysis:** determine whether hierarchical routing alone, or a smaller combination of enhancements, preserves most of the gain.

8 Conclusion

The project demonstrates that **adaptive multimodal routing can improve customer segmentation geometry**, but the path to that result is as important as the result itself. The initial MoE failed because simple learned gates and a reconstruction-oriented objective did not create useful specialization. The improved hierarchical architecture, trained with cluster-aware and diversity-oriented mechanisms and evaluated across multiple seeds, exceeded the strongest static-fusion baseline.

Research question	Answer	Evidence
Does adaptive routing improve on static fusion?	**Conditionally yes**	Phase 10: no. Phase 13 hierarchical MoE: 0.590873 versus 0.557928
Are simple learned gates sufficient?	**No**	Best Phase 10 MoE trails static fusion by 0.183403 silhouette
Is static fusion a weak baseline?	**No**	Text + temporal fixed-weight fusion is strong and highly stable
Does hierarchical MoE dominate every unimodal model?	**Not established**	Temporal-only silhouette is 0.005185 higher; other metrics favor hierarchical MoE
Is availability-aware masking necessary?	**Yes**	Indicator-free text + temporal fusion loses 0.276788 silhouette and 0.137750 ARI
Does better geometry imply better retention?	**No**	Temporal-only has strong geometry but weak retention evidence; RFM leads retention lift
Is the hierarchical MoE ready for production?	**Not yet**	Retention, actionability, aligned benchmarking and new-holdout validation remain open

Final assessment

The research hypothesis is **supported for sophisticated hierarchical routing against static fusion**. The hierarchical MoE is the strongest multimodal candidate in the supplied results, with silhouette 0.590873, DBI 0.638034, CH 19,223.176 and stability ARI 0.999912. KMeans RFM remains the strongest reported retention model at 3.756427× corrected lift. The next decisive milestone is to demonstrate that the hierarchical MoE's geometric advantage produces reliable, interpretable and useful downstream segments.

8.1 Phase-level synthesis

Phase	Purpose	Key report-worthy result
6	Legacy baselines	KMeans RFM: 0.5314 silhouette and 3.756427× corrected retention lift
7	Text representation	22,535 customers represented; zero cutoff violations and non-finite embeddings
8	Temporal representation	57,510 customers represented from 59,427 pre-cutoff events
9	Static fusion	Text + temporal fixed-weight fusion: 0.557928 silhouette
10	Initial MoE	Best MoE: 0.374525; adaptive hypothesis rejected for simple gates
11	Ablations	Simple learned gates show no significant retention separation
12	Fusion research question	Availability-aware masking is essential; indicator-free fusion degrades
13	Improved routing	**Hierarchical MoE exceeds best static fusion on silhouette, DBI and CH**

Appendix A Metric definitions

Metric	Definition and interpretation
Silhouette score	For a customer, compares mean within-cluster distance a(i) with the nearest-other-cluster distance b(i): s(i) = [b(i) − a(i)] / max[a(i), b(i)]. Values range from −1 to 1; higher values indicate better cohesion and separation.
Davies–Bouldin index	Average similarity between each cluster and its most similar other cluster, incorporating within-cluster scatter and centroid separation. Lower values indicate better-separated clusters.
Calinski–Harabasz index	Ratio of between-cluster dispersion to within-cluster dispersion. Higher values indicate denser, better-separated clusters; absolute values are not directly comparable across unrelated datasets or representations.
Adjusted Rand Index for stability	Agreement between cluster assignments across repeated fits, corrected for chance. Values near 1 indicate highly reproducible assignments under the tested procedure, not necessarily predictive or external validity.
Maximum cluster retention	The highest future retention rate observed among a model's clusters in the evaluation window. Small clusters can produce unstable maxima and should be interpreted with cluster sizes and intervals.
Corrected retention lift	Maximum cluster retention divided by the test retention baseline. The test baseline is 108 / 8,621 = 1.252755%; a lift of 3.756427× corresponds to 4.705882% maximum cluster retention.
Wilson 95% interval	A binomial proportion interval with better behavior than a simple normal approximation for small samples or proportions near zero. It communicates uncertainty in a cluster retention estimate.
Cohen's h	Effect size for a difference between two proportions based on arcsine transformation. It complements statistical significance, especially when event counts are small.
Omnibus p-value	Evidence against the null hypothesis that retention is equal across clusters. It does not identify which pairs differ and should be interpreted with prior test exploration and multiplicity in mind.

A.1 Key derived comparisons

Comparison	Calculation	Result
Hierarchical MoE vs best static fusion	0.590873 − 0.557928	**+0.032945 silhouette**
Relative silhouette gain vs static fusion	0.032945 / 0.557928	**+5.9%**
Hierarchical MoE vs temporal-only silhouette	0.590873 − 0.596058	**−0.005185 silhouette**
Hierarchical MoE vs Phase 10 best	0.590873 − 0.374525	**+0.216348 silhouette**
Indicator-free masking effect	0.319270 − 0.557928	**−0.276788 silhouette**
Text availability gap	100% − 39.2%	**60.8% without text**

End of report • Analysis date: 25 October 2026 • Project status: Phase 13 complete